

Life Star Academy By Sir Afzal

Question No: 16 (Marks: 1) - Please choose one

Is a data structure that can grow easily dynamically at run time without having to copy existing elements?

- ▶ Array
- ▶ **List**
- ▶ Both of these
- ▶ None of these

▶ Cs301 MidTerm Object File:07



▶ (Lectures from 1 to 23)



▶ **Written By: Sir Ray Afzal**



▶ **Whatsapp Number: 03016812299**



▶ **YouTube Channel: Life Star Academy**



Life Star Academy By Sir Afzal

Question No: 1 (Marks: 1) - Please choose one

Which one of the following statement is NOT correct .

- ▶ **In linked list the elements are necessarily to be contiguous** [Click here for detail](#)
- ▶ In linked list the elements may locate at far positions in the memory
- ▶ In linked list each element also has the address of the element next to it
- ▶ In an array the elements are contiguous

Question No: 2 (Marks: 1) - Please choose one

In a program a reference variable, say x, can be declared as

- ▶ **int &x ;** [Click here for detail](#)
- ▶ int *x ;
- ▶ int x ;
- ▶ None of the given options

Question No: 3 (Marks: 1) - Please choose one

Linked lists are collections of data items "lined up in a row" , insertions and deletions can be made only at the front and the back of a linked list.

- ▶ True
- ▶ **False** [Click here for detail](#)

Question No: 4 (Marks: 1) - Please choose one

A Linear Data Structure is the data structure in which data elements are arranged in a sequence or a linear list. Which of the following is Non Linear Data Structure?

- ▶ Arrays
- ▶ LinkLists
- ▶ **Binary Search Trees** [Click here for detail](#)
- ▶ None of these

Question No: 5 (Marks: 1) - Please choose one

A queue where the de-queue operation depends not on FIFO, is called a priority queue

- ▶ False
- ▶ **True** (Page 101)

Life Star Academy By Sir Afzal

Question No: 6 (Marks: 1) - Please choose one

Which one of the following statements is correct?

- ▶ **Array size is fixed once it is created** [Click here for detail](#)
- ▶ Link List size is fixed once it is created.
- ▶ Binary Search Tree size is fixed once it is created
- ▶ AVL Tree size is fixed once it is created

Question No: 7 (Marks: 1) - Please choose one

Which one of the following is correct about pointers?

- ▶ They always point to different memory locations
- ▶ They may point to a single memory location
- ▶ The address of two pointer variables is same
- ▶ None of these

Question No: 8 (Marks: 1) - Please choose one

Which of the following abstract data types are NOT used by Integer Abstract Data type group?

- ▶ short
- ▶ Int
- ▶ **float** [Click here for detail](#)
- ▶ long

Question No: 9 (Marks: 1) - Please choose one

The operation for adding an entry to a stack is traditionally called :

- ▶ add
- ▶ append
- ▶ insert
- ▶ **push** (Page 53)

Question No: 10 (Marks: 1) - Please choose one

The operation for removing an entry from a stack is traditionally called:

- ▶ delete
- ▶ peek
- ▶ **pop** (Page 53)
- ▶ remove

Question No: 11 (Marks: 1) - Please choose one

We can add elements in QUEUE From _____

- ▶ Front
- ▶ **Rear** (Page 91)
- ▶ From Both Rare and Front
- ▶ None of these

Life Star Academy By Sir Afzal

Question No: 12 (Marks: 1) - Please choose one

The difference between a binary tree and a binary search tree is that ,a binary search tree has

► **two children per node whereas a binary tree can have none, one, or two children per node** [Click here for detail](#)

- in binary search tree nodes are inserted based on the values they contain
- in binary tree nodes are inserted based on the values they contain
- none of these

Question No: 13 (Marks: 1) - Please choose one

Suppose n is the number of nodes in a complete Binary Tree then maximum steps required for a search operation are,

► **$\log_2(n+1) - 1$** (Page 139)

- $\log_2(n+1)$
- $\log_2(n) - 1$
- $\log_2(n)$

Question No: 14 (Marks: 1) - Please choose one

The following is a segment of a C program.

```
int pqr(BinaryNode t)
{ if (t == null )
return -1;
else
return 1+max(pqr(t.left),pqr(t.right)) }
```

Identify, what the above program intend(s) to do?

- Compute the height of a binary tree using an in-order traversal
- Compute the height of a binary tree using a pre-order traversal
- Compute the depth of a binary tree using a pre-order traversal
- Compute the depth of a binary tree using a post-order traversal

Question No: 15 (Marks: 1) - Please choose one

Consider the following infix expression:

$3 + 5 * 6 - 7 * (8 + 5)$

Which of the following is a correct equivalent expression(s) for the above?

- $3 \ 6 \ 5 + * \ 7 \ 5 \ 8 + - *$
- $3 \ 6 \ 5 \ 7 \ 5 \ 8 + * + - *$
- $3 \ 5 \ 6 + * \ 7 \ 8 \ 5 + - *$
- **$3 \ 5 \ 6 * + \ 7 \ 8 \ 5 + * -$**

Question No: 16 (Marks: 1) - Please choose one

An array is a group of consecutive related memory locations.

- **True** [Click here for detail](#)
- False

Question No: 17 (Marks: 1)

Life Star Academy By Sir Afzal

Is this a correct statement? Give answer in Yes or No.

A node cannot be deleted, when the node to be deleted has both left and right subtrees.

False----- No, it can be deleted.

Question No: 18 (Marks: 1)

Deleting a leaf node in binary search tree involves setting_____pointer/s of that node's parent as null.

- 1
- 2
- 3
- 4